

Microinteractions for supporting grasp tasks through usage of spare attentional and motor resources

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ABSTRACT

This paper concludes the nine months progress of my Ph.D. project. It specifies its research focus on investigating microinteractions – a sub-topic of HCI and gesture research – and on developing a frame work for applications, which use microgestures for supporting everyday tasks through invisible and context-aware appearing interface layers underneath object-grasping hands. In an expert study, I explore the motor limitations and opportunities of microgestures while grasping objects and valued manual dual-task scenarios by walking through three tasks, with usage of grasping probs. The study's outcome is a generic microgesture set for different grasp types and a collection of parameters that have a relevant effect on the choice of the grasping tasks. A further user study in progress is investigating the effect of grasped object, such as handheld devices, on the feasibility of microgestures. In this study users are asked to perform finger-tip and drags on the front and / or back side of a held device. For this user study I have used a two-sided touch-sensitive device by stacking 2 pads together to a sandwich-like prototype. This allows tracking users' finger gestures through camera as well as through front and touch screens. The outcome of the two mentioned studies will describe a design space for out-of-a-grasp microgestures. At the ECCE doctoral consortium I aim to present this design space and discuss how this can serve as a base for developing a framework for out-of-grasp microinteractions that are subtasks of grasping tasks. The microinteractions will be developed to support the grasp tasks regarding their perceived ergonomic and hedonic qualities.

Keywords

Dual-task, multitask, interaction style, gesture

Research focus

Human actions in the real world are usually a set of in-parallel solved tasks and people have the ability for multitasking with regards to motor and attentional resource economics in lots of everyday situations. The human-computer interactions are normally designed as a separate task and synchronous tasks are understood as multitasking situations with two competitive tasks

regarding their motor and attentional resource requirements. Microinteractions are a promising technique to be done in parallel to manual actions with less competitive motor resource effort.

Ashbrook defines microinteractions as short-time interruptions of primary tasks for controlling mobile devices [1]. These interaction techniques have the beneficial potential of allowing mobile application control in parallel to ongoing everyday tasks. This could significantly enrich the quality of tasks that we could perform on-the-go and fundamentally alter the way we view ubiquitous computing [8]. In this context a microinteraction can be designed as a support or subtask of an everyday task or contextual independent, like mobile communications.

I aim to investigate the potential of microinteractions, which are a subtask of a primary one. In this field several research projects have recently been conducted about development tracking interfaces [1, 2, 5, 8] and a few that are driven by the human side of interaction [3, 4, 12]. I see a research gap on the human-centric side of microinteractions, and I am motivated to investigate the design space of these interactions; mainly their affordances and constraints regarding usability aspects, such as ergonomic and hedonic qualities. This in a further step shall guide my research in developing a framework for sub-tasking microinteractions that support grasp tasks. Subtasks in this context are e.g. automotive control interactions like releasing the blinking light or opening the window while driving. Realizing these subtasks through microgestures allows their performance in parallel to steering a car by keeping the steering wheel fully grasped as figure x shows.

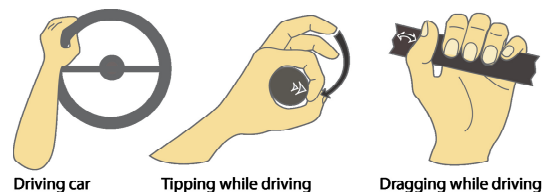


Fig. 1. Performable microgestures while steering a car with a palm grasp: Tipping fingers on the wheel or dragging it with the thumb.

I believe in the possibility that synchronous microinteractions allow supporting every-day tasking in an ergonomic and hedonic manner. Therefore my hypothesis is: Microinteractions have the potential to increase the perceived ergonomic and hedonic qualities of manual tasks if they are performed in parallel and contextual related.

Understanding microinteractions

For understanding microinteractions, I did an expert evaluation [13]. I asked sports- and physiotherapists for using props (as shown in figure 2) while performing given microgestures to evaluate their motor and attentional effort. But the experts were also asked for creating new gestures, which just came into their mind.

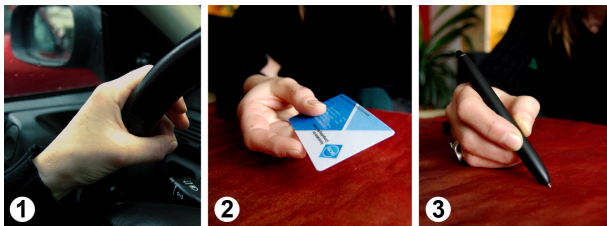


Fig. 2. The participants are testing the feasibility of finger gestures while (1) holding a steering wheel (2) targeting a cash card, and (3) drawing with a pen.

The gestures should be easily performable without interrupting the primary task through requiring high motor effort, and their performance itself should also just need none or low attention.

Human-computer interactions require (depending their design and the experience level of the user) more or less attention for some, but not all of their interaction stages that are for instance goal definition, finding the right interaction technique that would support reaching that goal, and performing the interaction technique. When we are using a mouse, this happens automatically because we practiced it so many times. But because users might not be used to perform the gestures we collected. For instance the vulcan salute, known from the Star Trek series, is not easy performable by many users. Therefore the study aimed to identify the level of attention that the pure gesture performance would require.

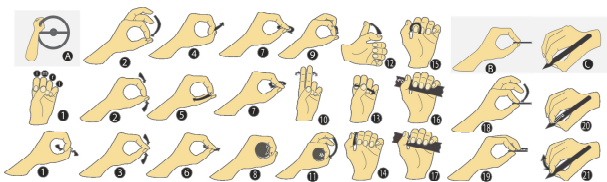


Fig. 3. The expert-defined and evaluated gesture set. The experts found 17 gesture types for the driving scenario (A). The card targeting scenario (B) and the stylus scenario (C) just contain 2 gesture types each. Most gesture types have several sub-types by performing them with different fingers (index, middle, ring, and little finger). Moreover the same gesture results in a different sub-type (e.g. touch, tab, or press), if it is performed with different acceleration or duration.

Resulting from the expert interviews I developed a taxonomy (see figure 3) for classifying these gestures according to their use cases and assessed their ergonomic and attentional attributes, focusing on their

primary task compatibility. I defined 21 gestures, which allow microinteractions within manual dual task scenarios. In expert interviews I evaluated the gesture's level of required motor or attentional resources under the constraint of stable primary task performance.

I asked the experts also for explaining issues that limit the motor gesture performance or that could require more attention.

The expert answers showed that anatomic aspects affect the gesture feasibility (see Fig. 4). To move the index finger separately is for the majority of the users easy doable. Moving the ring finger without moving the "neighbour" fingers is much harder. Users are still able to do this, but usually they have to pay a lot of attention on this movement.

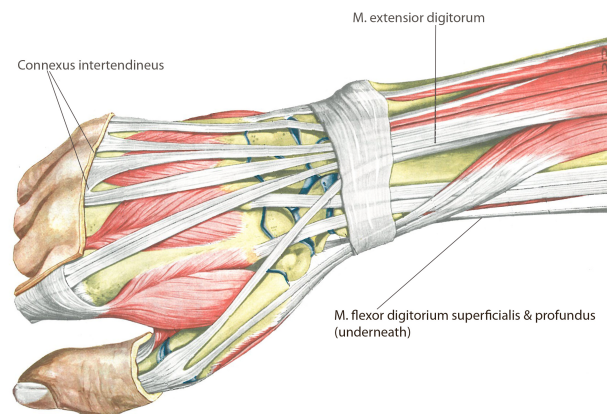


Fig. 4. shows the anatomic connection between the fingers that is responsible for the separation problem of the ring finger. Fig. 4 is a simplification of a figure in Spalteholz' Anatomy of Human [9].

This study was supervised by three researchers (one psychologist and two computer scientists) of the research group I am associated with at the Quality and Usability Laboratories of TU Berlin and that are named as co-authors.

Designing microinteractions

Gestural interfaces miss affordances and constraints which are readily provided by other interfaces, such as graphical and tangible [6, 7]. In particular, it is difficult to let users know what they are able to do, what they are currently doing or what they just have done. Because of this, gestural interfaces and in particular microgestures are not to be understood as a replacement for other kinds of interfaces, but rather as enabling novel ways of interaction. There are still many open questions to be answered, especially regarding interaction opportunity and feedback representation.

Our taxonomy is mainly investigating ergonomic interaction opportunities of microinteractions and can be used as a basis for designing microinteraction techniques for manual dual-task scenarios: First, the scenario has to be analyzed for defining the limits and requirements for microgestures. Then, and by having a look at the formal structure of the chosen gestures, a gesture set can be defined. Lastly, a decision about the

sensory and tracking requirements of the hardware can be made.

For a formal scenario design, we propose two synergetic strategies to guide the scenario design: the handling of motor resources as well as of attentional ones.

The selection of the primary (main-) and the secondary (independent or sub-) task is reasoned by the idea that automatically performed tasks require low attention [11]. This allows in parallel tasking and means, even when our primary and secondary tasks use equal modalities by requiring tactile feedback and kinesthetic self-awareness, an automatically performed task allows paying attention on simultaneously performing secondary tasks such as microinteractions. These circumstances allow performing two tasks in parallel (table 2, column 4). The example steering a car represents an automatically performed and adaptively aware task that in normal driving situations (holding or changing the lane) not needs much attention. Controlling the navigation system by hand gestures could be a secondary one using available attentional resources for processing.

Analogue to this idea of allowing two task performances in parallel, there are also spare potentials in hand's motor resources, even when the hand is involved in a grasping task. We mentioned that most manual tasks not require all fingers (table 1, column 4). For instance, steering a car is still possible when the fingers are moving. The primary task defines the usage of motor resources as well as free potentials and available hand motor skills that can be used for simultaneous tasks. The grasp type that is performing the primary task (palm, pad, side) defines the motor resources which are used in the primary task (see Table 1, column 3). Our taxonomy identifies microinteractions executable in parallel based on free motor resources (Table 2, column 1-3) and allows to create a hand gesture set for commanding the secondary task. The hand or microgesture design is driven by identifying fingers that are not necessarily or not permanently involved in the grasp task.

We aim using the mentioned spare attentional and motor resources for designing microinteractions and in a more abstract manner for re-thinking every-day grasp-based tasks.

Enabling dual tasking with mobile devices while continuing a main task is actually not limited to the scenarios, we have chosen for the experiment. Our concept is more general and investigating the opportunities that come with novel coupling tasks is very promising. Microinteractions that use new sensors offer the possibility of ambient interface access. The focus of both our expert study and our further research is upon microgestures that are performable while a manual primary task is continued. In this study we have chosen exemplary tasks for the three main grasp types of the human hand. Because in the first step we identified ergonomically feasible hand gestures, we

created a generic gesture set for designing such interactions. Finger gestures seem to be a beneficial approach for this, and they can serve as an interface for sub- or in parallel tasking for a wide range of every-day tasks.

Research plan and status quo

My Ph.D project is planned to consist of three parts and I am currently working on the first one. This part investigates the fundamental understanding of microinteractions. Within the second part, I will – based on the expertise of part 1 – develop novel microinteraction techniques and evaluate them regarding their usability, such as ergonomic and hedonic qualities. The third part will be dedicated to develop a framework for microinteractions that will serve as seamless sub-tasks to support in parallel continued grasp-based tasks.

My first experiment [15] serves as an initial analysis basis for understanding microinteractions and will be continued to generate a fundamental description of the design space for out-of-grasp microgestures. In this initial study, attributes that have a relevant effect on the gesture performance are defined. These are the grasp type, the grasped objects, ergonomic and bio-mechanic aspects, as well as the attention that is required for the pure gesture performance. The benefit of microinteractions on the usability of the grasp tasks was valued by experts as useful augmentation for many grasp-based tasks but they also brought up the fact that the degree of benefit might be influenced by pragmatic aspects, such as the task length and by ergonomic parameters of the grasped object.

Guided by the conclusion of the expert study, I currently plan a user study to investigate how users grasp an object like a pad and the effect of physical object attributes (use, size, and shape) on the way users grasp it and on their finger gesture feasibility. The expected outcome of this study is a deeper understanding about users' naturally chosen grasp for holding a pad (e.g. iPad) while performing certain device interactions as well as an evaluation of users' ergonomic skills of performing gestures while grasping the device. To measure the limitations of the grasp performance I will choose certain tasks that have to be done under different conditions (with one or two hands / with or without releasing other fingers). A generic set of finger gestures that are performable by a grasping hand are given by the Taxonomy of Microinteractions [15]. To measure the specific gesture limitations while grasping an object, we ask users to perform guided gestures while holding an iPad sandwich.

Contribution and expected benefit of the doctoral consortium

Within the student consortium I would appreciate presenting my previous research [13], [14] and the results of my first study [15] shortly for explaining my current work in greater detail.

I would appreciate getting feedback about my applied methodology of doing expert interviews for evaluating the ability of users to perform gestures.

Another question I would like to discuss is how to handle the fact that the prototype of the planned user study is much heavier than the actual products will be, and also that the time of usage might be much shorter as in many real use cases. Both factors will influence the experiments measurements and limit their validity.

Finally, because I aim to develop a framework for microinteractions, I would appreciate advices about how to get valid measurements that are scalable for various grasp-based task types and for grasped objects with different parameters.

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